Statistics & Data Analysis

ClipSAT Printable Practice Suite (clipsat.org) • Topic 11.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Review: The real number system

The function $f(x) = a \cdot b^x$ passes through the points $(1, 12)$ and $(3, 108)$. Determine the value of a and the value of b, where $a > 0$ and $b > 0$.

2 [Foundational] • Target: Review: Integer operations

The first four terms of an arithmetic sequence are $7, 11, 15, 19, \ldots$. Which expression represents the nth term, a_n, of the sequence for all integers $n \geq 1$?

3 [Foundational] • Target: Review: Operations with rational numbers

An arithmetic sequence is defined by its first term $a_1 = 7$ and its common difference d. The sum of the first n terms of this sequence is given by $S_n = \frac{n}{2}(2a_1 + (n-1)d)$. If the sum of the first 10 terms is $S_{10} = 115$, find the value of the common difference d.

4 [Foundational] • Target: Review: Exponents and radicals

The first three terms of an arithmetic sequence are given by $a_1 = 5$, $a_2 = 8$, and $a_3 = 11$. What is the 10th term of this sequence?

5 [Foundational] • Target: Review: Laws of exponents

Consider the arithmetic sequence where the first term is 5 and the common difference is 3. What is the 4th term of this sequence?

6 [Foundational] • Target: Units resulting from a formula or graph

Simplify the expression: $\frac{2^5 \cdot 2^3}{2^4}$.

7 [Foundational] • Target: Review: The real number system

Simplify the expression: $(3x^2y^3)^2$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Review: Integer operations

A line passes through the points $(2, 3)$ and $(6, 11)$. What is the slope of this line?

9 [Intermediate] • Target: Review: Operations with rational numbers

What is the solution to the equation $2^{x+1} = 32$?

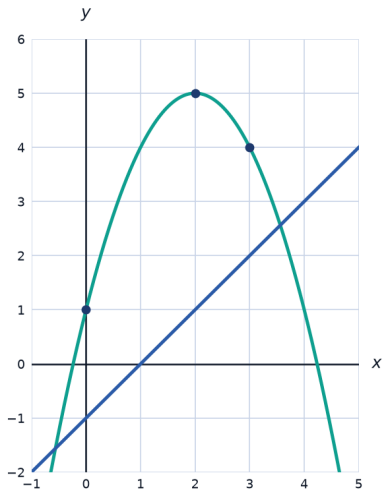
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10 [Intermediate] • Target: Review: Exponents and radicals

Simplify the expression: $(a^3)^2 \cdot a^4$

11 [Intermediate] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(5)$.

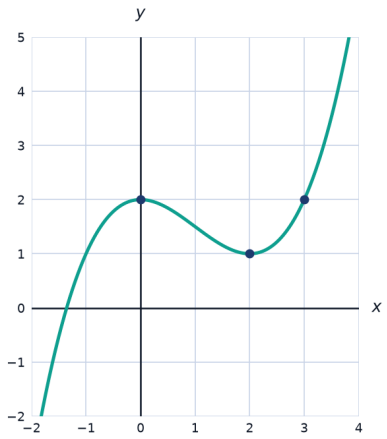


12 [Intermediate] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

13 [Intermediate] • Target: Review: Exponents and radicals

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(2)$.



14 [Intermediate] • Target: Review: Exponents and radicals

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 6.01 Systems & Inequalities

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$.

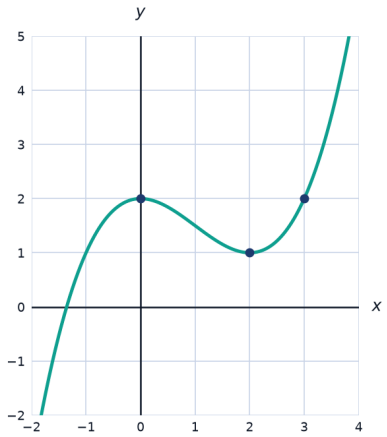
What is the arithmetic mean? Show your complete algebraic work.

16 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 7.01 Sequences & Series

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(3)$.

17 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 8.01 Factoring Polynomials

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.



18 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 9.01 Radicals & Exponents

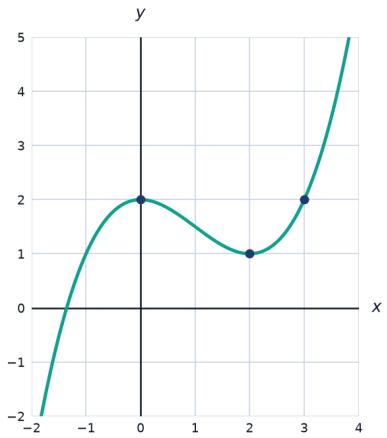
For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(4)$.

19 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 1.01 Linear Equations & Lines

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

20 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 10.01 Rational Expressions

For the model $f(x) = 5x + 6$ in review: exponents and radicals, evaluate $f(5)$.

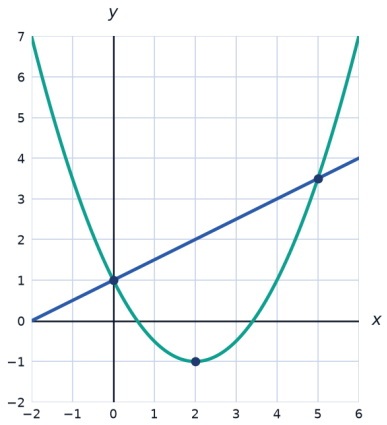


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Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(2)$.



a. 7

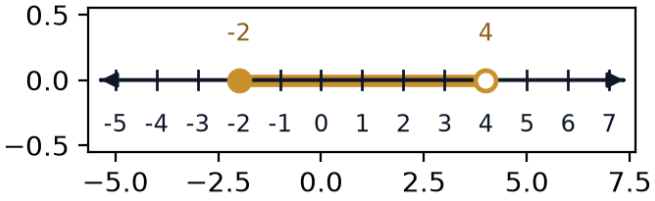
b. 9

c. 4

d. 11

22 [Foundational] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.



a. $x = 2$ and $x = 5$

b. $x = 2$ and $x = 4$

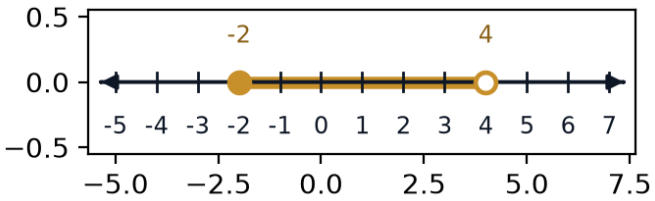
c. $x = -2$ and $x = -4$

d. $x = 6$ and $x = 8$

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23 [Foundational] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(5)$.



a. 13

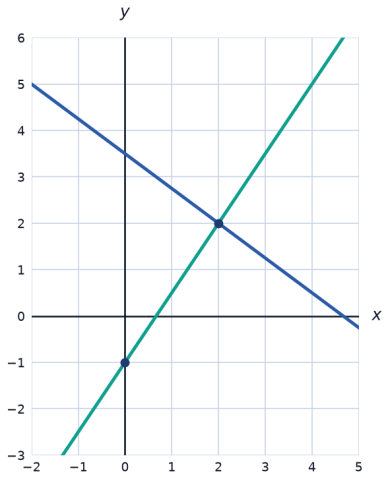
b. 15

c. 10

d. 17

24 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(2)$.



a. 10

b. 13

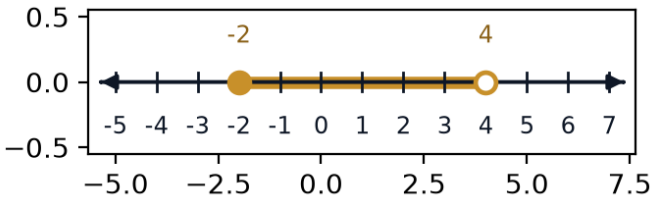
c. 6

d. 14

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25 [Foundational] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.



a. $x = 2$ and $x = 6$

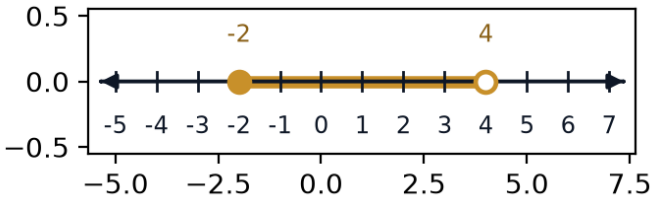
b. $x = 2$ and $x = 5$

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

26 [Foundational] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(5)$.



a. 19

b. 22

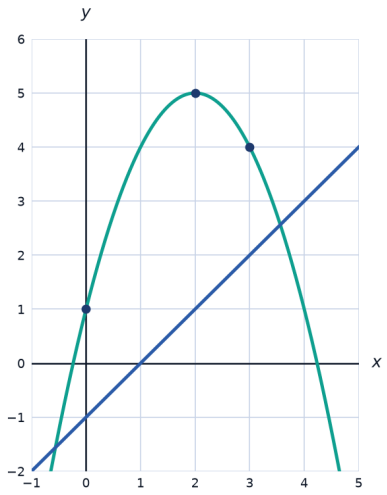
c. 15

d. 23

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27 [Foundational] • Target: Review: Operations with rational numbers

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(2)$.



a. 13

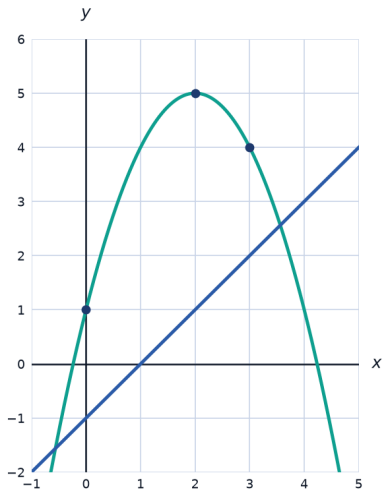
b. 17

c. 8

d. 17

28 [Foundational] • Target: Review: Operations with rational numbers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



a. $x = 2$ and $x = 7$

b. $x = 2$ and $x = 6$

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

30 [Intermediate] • Target: Review: The real number system

A data set recorded for review: the real number system consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5

d. 18.5

31 [Intermediate] • Target: Review: The real number system

For the model $f(x) = 2x + 3$ in review: the real number system, evaluate $f(3)$.

a. 9

b. 11

c. 6

d. 13

32 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

33 [Intermediate] • Target: Review: The real number system

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

a. $x = 4$ and $x = 7$

b. $x = 4$ and $x = 6$

c. $x = -4$ and $x = -6$

d. $x = 10$ and $x = 24$

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34 [Intermediate] • Target: Review: Integer operations

A data set recorded for review: integer operations consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5

d. 19.5

35 [Intermediate] • Target: Review: Integer operations

For the model $f(x) = 3x + 4$ in review: integer operations, evaluate $f(3)$.

a. 13

b. 16

c. 9

d. 17

36 [Intermediate] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ and $x = 6$

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

37 [Intermediate] • Target: Review: Integer operations

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ and $x = 7$

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 9.01 Radicals & Exponents

A data set recorded for review: operations with rational numbers consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5

d. 20.5

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39 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 1.01 Linear Equations & Lines

For the model $f(x) = 4x + 5$ in review: operations with rational numbers, evaluate $f(3)$.

a. 17

b. 21

c. 12

d. 21

40 [Challenging] • Target: Review: Operations with rational numbers • Cross-Lesson: 10.01 Rational Expressions

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ and $x = 7$

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

41 [Challenging] • Target: Review: Exponents and radicals • Cross-Lesson: 2.01 Quadratic Functions

A data set recorded for review: exponents and radicals consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5

d. 21.5

42 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 3.01 Exponents & Polynomials

A data set recorded for review: laws of exponents consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5

d. 22.5

43 [Challenging] • Target: Review: Laws of exponents • Cross-Lesson: 4.01 Functions & Transformations

For the model $f(x) = 6x + 7$ in review: laws of exponents, evaluate $f(3)$.

a. 25

b. 31

c. 18

d. 29

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44 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

A data set recorded for units resulting from a formula or graph consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

45 [Challenging] • Target: Units resulting from a formula or graph • Cross-Lesson: 6.01 Systems & Inequalities

For the model $f(x) = 2x + 8$ in units resulting from a formula or graph, evaluate $f(3)$.

a. 14

b. 16

c. 6

d. 18

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